Machine Learning: short introduction (trees and ANN), part 3

Sergey Korpachev on behalf of Dépôt

Outline

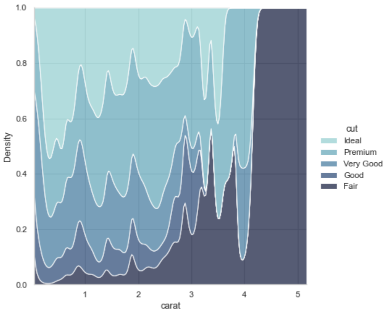
- Machine learning (ML)
- Data
- Features in ML
- ML pipeline
- Linear model
- Decision tree
- Neural network
- Summary

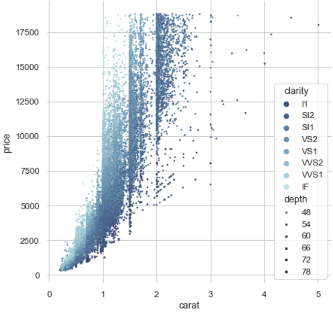
Modelling nonlinearities

Modelling nonlinearities

They exist

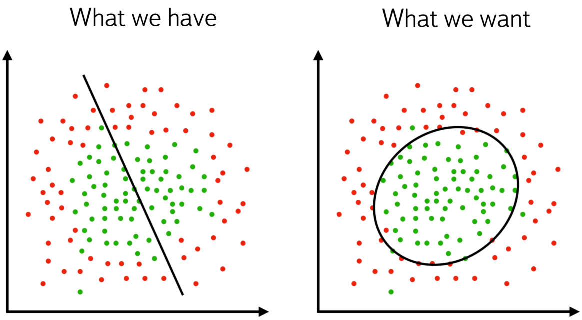
seaborn illustrations





Modelling nonlinearities

Linear models

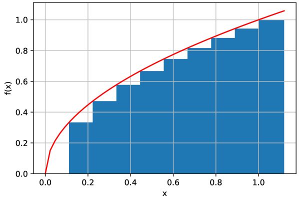


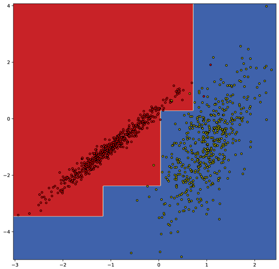
Linear models can't simply describe complex nonlinear data

Modelling nonlinearities

Trees

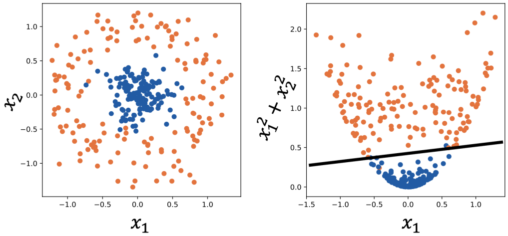
- (Ensembles of) Trees were designed to approximate nonlinearities and are **pretty good** in it + they are **fast and interpretable**
- But they are just "brute-force" algorithms – **don't infer symmetries** in data by design
- **Ad-hoc, cut-based and piecewise approximations** of data at hand + not differentiable and smooth





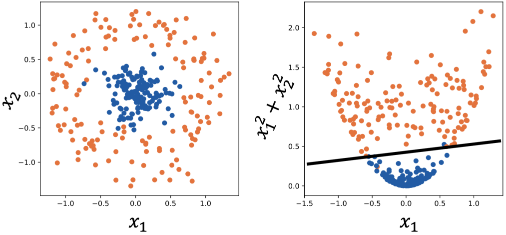
Modelling nonlinearities

Feature engineering



- But sometimes we know *a priori* that there are transformations simplifying the problem $\to$ even linear model can do the job
- However, this **feature engineering** is non-trivial, requires domain knowledge and is time-consuming

Modelling nonlinearities



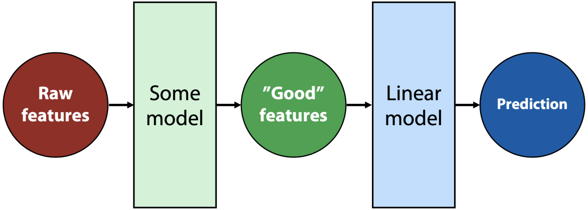
- But sometimes we know *a priori* that there are transformations simplifying the problem $\rightarrow$ even linear model can do the job
- However, this **feature engineering** is non-trivial, requires domain knowledge and is time-consuming

What if we design a model which could **automatically** feature-engineer itself?

Neural Network

Neural Network — Automating FE

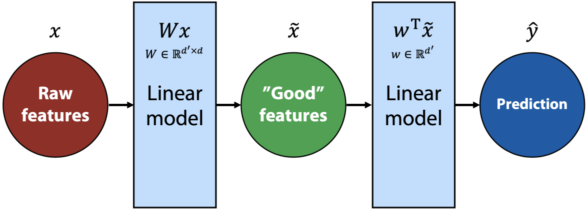
NN illustrations from
ML in HEP 2020



- Let's use a **simple linear model** to solve our supervised problem
- Add a block to a linear model which will automatically **generate new features** for it
- Two blocks would work together as a **single model** $\Rightarrow$ their parameters are updated simultaneously
- And **automatically**, by e.g. gradient descent (given their differentiability)

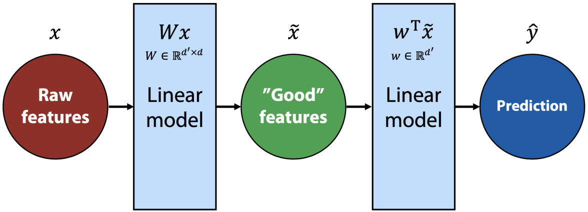
Neural Network

Automating FE



- Would a linear model work as a feature generating model?

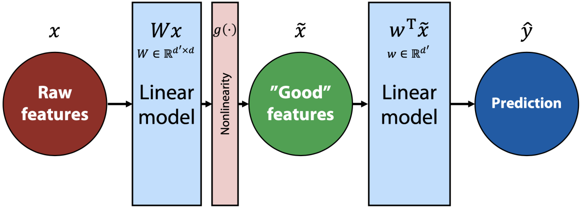
Neural Network — Automating FE



- Would a linear model work as a feature generating model? No
- $\hat{y} = w^T \tilde{x} = w^T(Wx) = (w^T W)x = w'^T x \Rightarrow$ it is still a linear model

➔ Input feature space has not changed, only the model weights

Neural Network — Automating FE

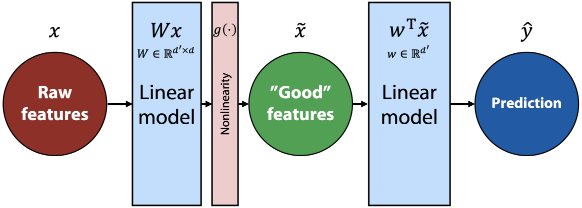


- Let's then introduce **nonlinearity** to our model

→ $\hat{y} = w^T\tilde{x} = w^T g(Wx)$,
where $g(\cdot)$ – some nonlinear scalar function (applied elementwise)

Neural Network

Automating FE

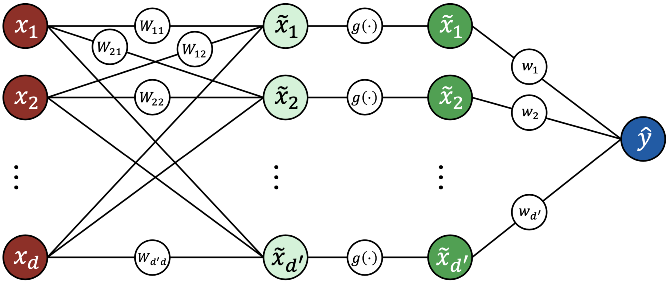


- Let's then introduce **nonlinearity** to our model

→ $\hat{y} = w^T \tilde{x} = w^T g(Wx)$,
where $g(\cdot)$ – some nonlinear scalar function (applied elementwise)

→ This is the simplest example of a **neural network**

Neural Network — Architecture

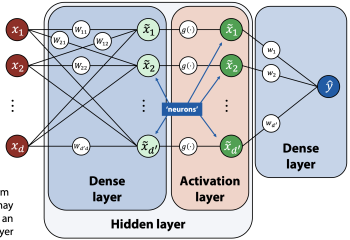


Essentially, NN is just a **composite function** that maps a set of X to a set of Y

$$\hat{y} = w^T \tilde{x} = w^T g(Wx)$$

Neural Network — Terminology

Feed-forward network:



Note: the term "activation" may also stand for an output of a layer

- Brown nodes $x_1, x_2, \ldots, x_d$ – features from an **input layer**
- Green nodes $\tilde{x}_1, \tilde{x}_2, \ldots, \tilde{x}_{d'}$ – **neurons** from a **hidden layer**
- Blue node $\hat{y}$ – neuron from an **output layer**
- Straight lines (edges) between neurons – **weights** w_{ij}
- $g(\cdot)$ – nonlinear **activation** function, e.g. sigmoid $\sigma(\cdot)$
- Important: each neuron has additional **bias** b associated to it and added to other inputs (not illustrated)

Training

Training

How to train?

- Since NN fundamentally is a parametrized differentiable model we can optimise the loss function and use **gradient descent** to train it:

$$\omega_{k+1} \leftarrow \omega_k - \eta \cdot \nabla Q(\omega_k)$$

- But gradients are hard to derive analytically – writing down all the derivatives is tough and tedious (especially for large NN)
- Note that NN is just a **composite model** $\Rightarrow$ can use **chain rule** for differentiating it

- Computing derivative of a "base" function is simple $\Rightarrow$ decompose composite function into a set of base ones and differentiate them one by one
- Let's recall the rule:

$$\frac{\partial f\big(t(x)\big)}{\partial x} = \frac{\partial f(t)}{\partial t} \cdot \frac{\partial t(x)}{\partial x}$$

Exercise: can you compute derivative of sigmoid function by decomposing it into base functions?

$$\sigma(x) = \frac{1}{1+e^{-x}}$$

Training — Chain rule

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Exercise: can you compute derivative of sigmoid function by decomposing it into base functions?

$$\sigma(x) = \frac{1}{1 + e^{-x}}$$

$$\sigma(x) = f(g(h(p(x)))), \quad f(z) = \frac{1}{z}, \quad g(z) = 1 + z, \quad h(z) = e^z, \quad p(z) = -z$$

Training

Chain rule

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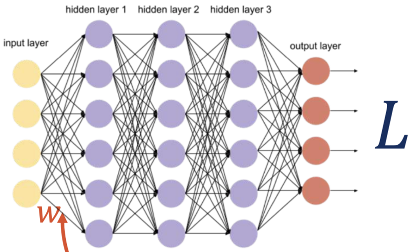
$$\sigma'(x) = \frac{\partial f}{\partial g} \cdot \frac{\partial g}{\partial h} \cdot \frac{\partial h}{\partial p} \cdot \frac{\partial p}{\partial x} = -\frac{1}{g^2} \cdot 1 \cdot e^p \cdot (-1) = \ldots = \sigma(x) \cdot \big(1-\sigma(x)\big)$$

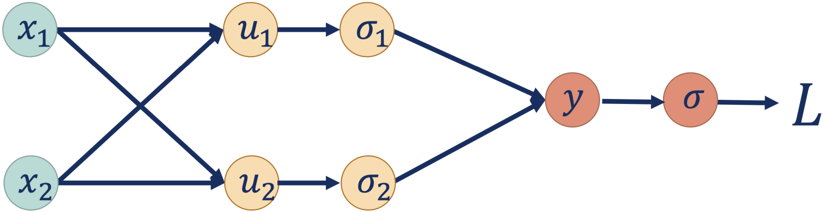
backprop illustrations from DMIA



$$w^{(t+1)} = w^{(t)} - \gamma_t \boxed{\frac{\partial L}{\partial w}}$$

- So how to find partial derivatives of loss function with respect to some weight?

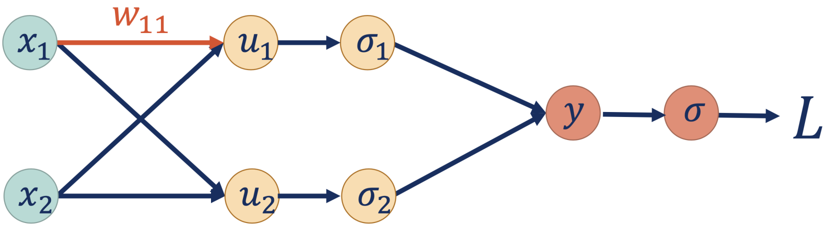
backprop illustrations from DMIA



- Let's consider a simplified neural network
- Represent it in the form of a **computational graph**

Training

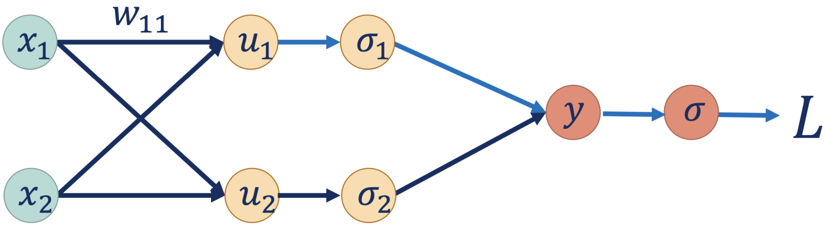
Backpropagation



$$w_{11}^{(t+1)} = w_{11}^{(t)} - \gamma_t \frac{\partial L}{\partial w_{11}}$$

Training

Backpropagation

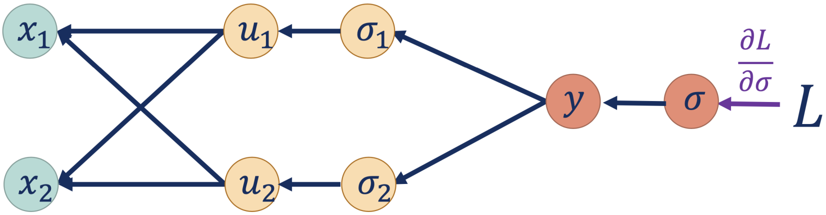


$$w_{11}^{(t+1)} = w_{11}^{(t)} - \gamma_t \frac{\partial L}{\partial w_{11}}$$

$$\boxed{\frac{\partial L}{\partial w_{11}} = \frac{\partial L}{\partial u_1}\frac{\partial u_1}{\partial w_{11}} = \frac{\partial L}{\partial u_1} x_1}$$

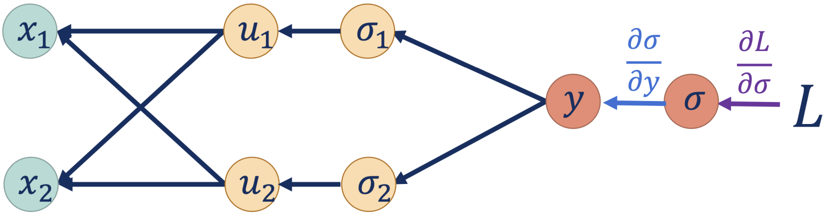
Training

Backpropagation



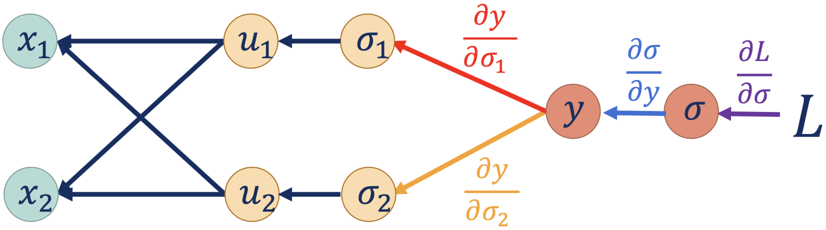
Training

Backpropagation



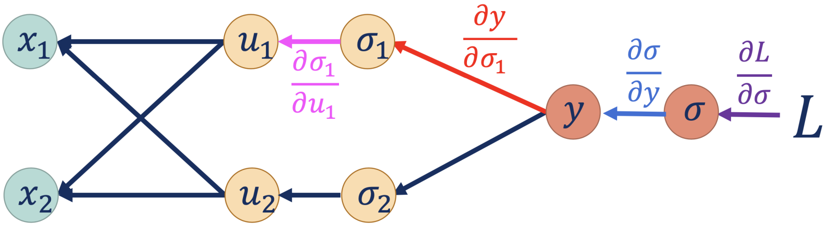
Training

Backpropagation



Training

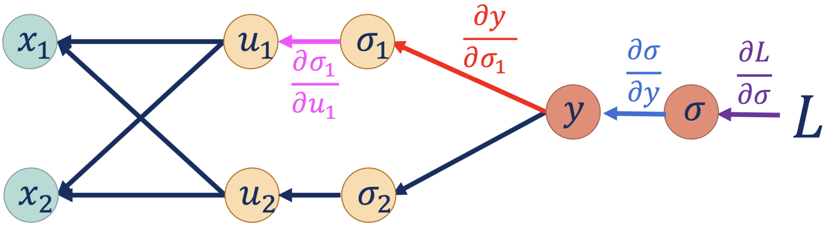
Backpropagation



$$\frac{\partial L}{\partial u_1} = \frac{\partial L}{\partial \sigma}\frac{\partial \sigma}{\partial y}\frac{\partial y}{\partial \sigma_1}\frac{\partial \sigma_1}{\partial u_1}$$

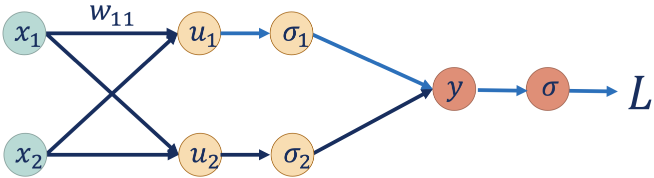
Training

Backpropagation

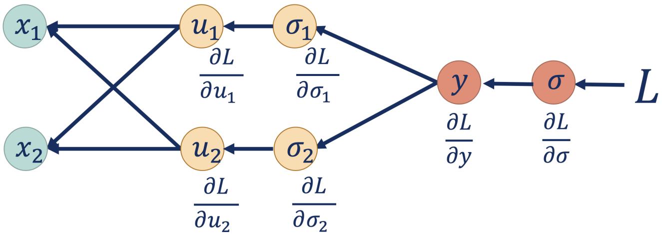


$$\frac{\partial L}{\partial u_1} = \frac{\partial L}{\partial \sigma}\frac{\partial \sigma}{\partial y}\frac{\partial y}{\partial \sigma_1}\frac{\partial \sigma_1}{\partial u_1}$$

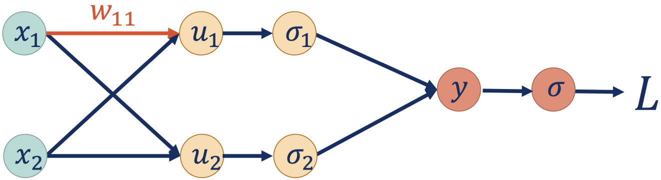
- this procedure is called **backpropagation**
- its idea is to **collect derivatives** at each step in the computational graph w/o recalculating them every single time for every weight



1. make **forward pass** through NN to calculate the output of each neuron and value of loss function



1. make **forward pass** through NN to calculate the output of each neuron and value of loss function
2. with **backward pass** go back through NN to calculate gradients for all weights



$$w_{11}^{(t+1)} = w_{11}^{(t)} - \gamma_t \frac{\partial L}{\partial w_{11}}$$

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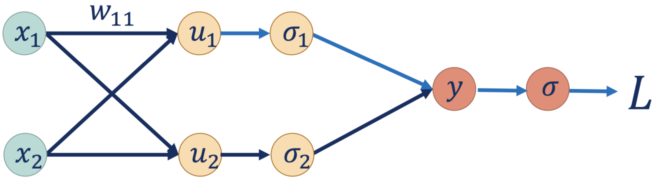


1. make **forward pass** through NN to calculate the output of each neuron and value of loss function
2. with **backward pass** go back through NN to calculate gradients for all weights
3. update weights with their gradients
4. repeat until convergence

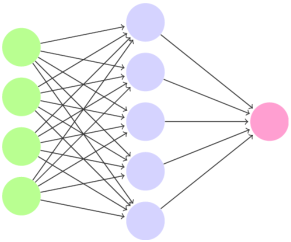
*one iteration (**epoch**) = forward and backward pass

Going deeper

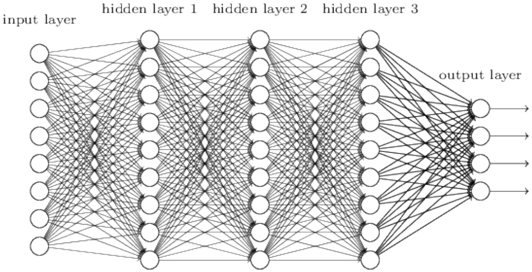
Going deeper

Universal approximation theorem

- Roughly speaking, any well-behaved function f can be approximated **arbitrarily close** with a 1-hidden layer NN, given wide enough hidden layer
- But in practice this is often not the case:
  - loss function is heavily non-convex
  - overfitting
  - not straightforward how to find this NN



- In practice stacking more layers generally **improves performance**

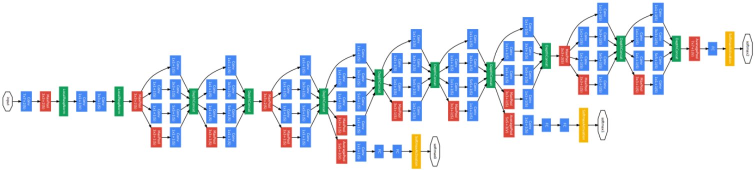


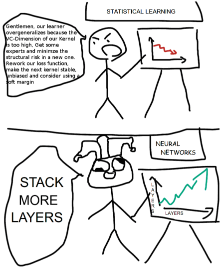
Going deeper

Stack more layers

Going deeper with convolutions

- In practice stacking more layers generally **improves performance**
- Much more layers...



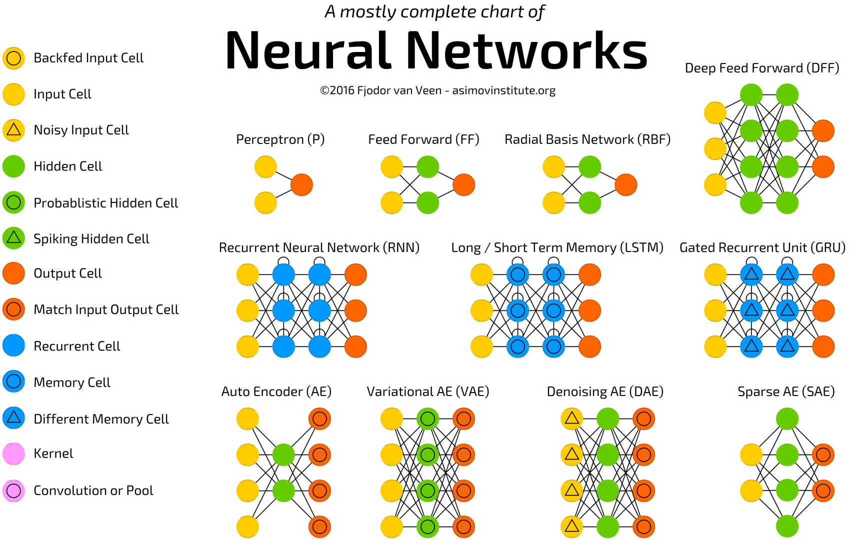


- In practice stacking more layers generally **improves performance**
- Much more layers...
- Which is reminiscent of the **brain structure**
- Signals travel through multiple areas of different organization
- This makes our perception system incredibly advanced in understanding reality

- In practice stacking more layers generally **improves performance**
- Much more layers...
- Which is reminiscent of the **brain structure**
- Signals travel through multiple areas of different organization
- This makes our perception system incredibly advanced in understanding reality

→ but there are some problems...

NN zoo


A mostly complete chart of
Neural Networks
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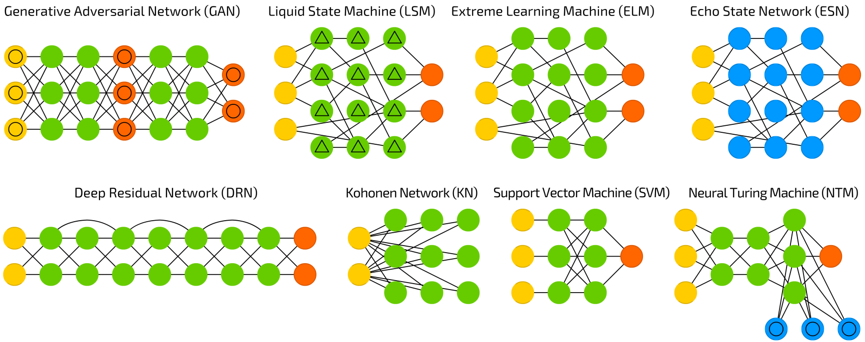
Backfed Input Cell
Input Cell
Noisy Input Cell
Hidden Cell
Probablistic Hidden Cell
Spiking Hidden Cell
Output Cell
Match Input Output Cell
Recurrent Cell
Memory Cell
Different Memory Cell
Kernel
Convolution or Pool
Perceptron (P)
Feed Forward (FF)
Radial Basis Network (RBF)
Deep Feed Forward (DFF)
Recurrent Neural Network (RNN)
Long / Short Term Memory (LSTM)
Gated Recurrent Unit (GRU)
Auto Encoder (AE)
Variational AE (VAE)
Denoising AE (DAE)
Sparse AE (SAE)

link



Generative Adversarial Network (GAN)
Liquid State Machine (LSM)
Extreme Learning Machine (ELM)
Echo State Network (ESN)
Deep Residual Network (DRN)
Kohonen Network (KN)
Support Vector Machine (SVM)
Neural Turing Machine (NTM)

Section summary

- Modelling nonlinearities
- Neural Network
  - automating feature engineering
  - architecture
  - terminology
- Training
  - chain rule
  - backpropogation
- Going deeper
  - universal approximation theorem
  - vanishing gradients
  - activation functions
  - weight initialisation
- Tackling overfitting
  - gradient descent modifications
  - weight regularization
  - dropout
  - batch normalisation
- NN zoo

Summary

From Wikipedia, the free encyclopedia

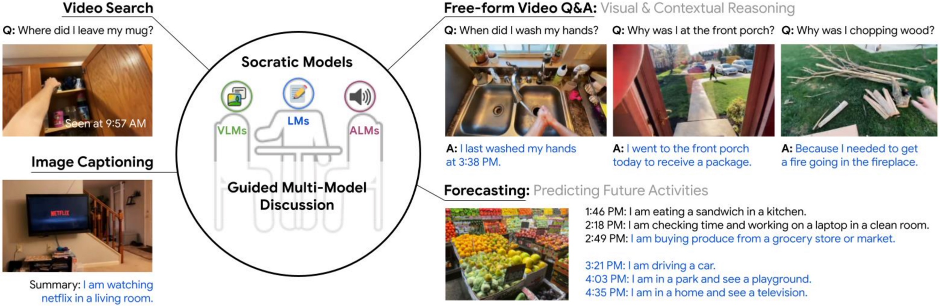
The **Socratic method** (also known as **method of Elenchus** or **Socratic debate**) is a form of argumentative dialogue between individuals, based on asking and answering questions.

In Plato's dialogue "Theaetetus", Socrates describes his method as a form of "midwifery" because it is employed to help his interlocutors develop their understanding in a way analogous to a child developing in the womb. The Socratic method begins with commonly held beliefs and scrutinizes them by way of questioning to determine their internal consistency and their coherence with other beliefs and so to bring everyone closer to the truth.

In modified forms, it is employed today in a variety of pedagogical contexts.

Summary

ML today

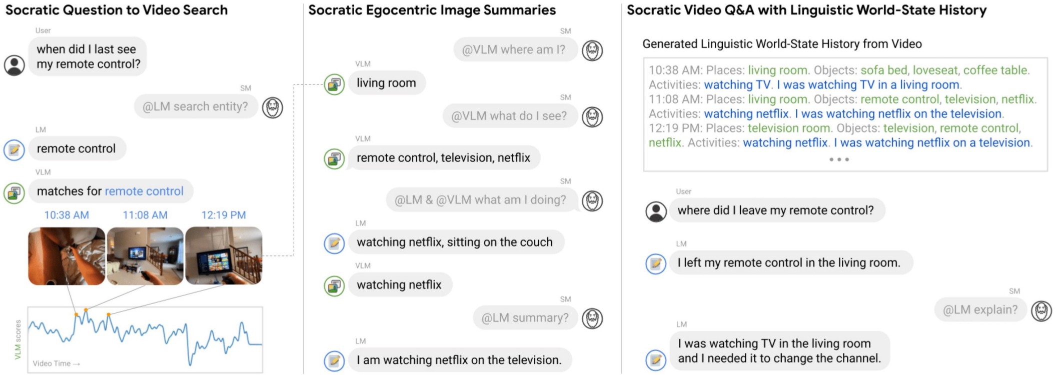


- No training
- Multimodality
- Commonsense
- Reasoning

Self-supervised generative \ metric pretraining

Language modeling

Zeng, Andy, et al. "Socratic models: Composing zero-shot multimodal reasoning with language." *arXiv preprint arXiv:2204.00598* (2022).



Zeng, Andy, et al. "Socratic models: Composing zero-shot multimodal reasoning with language." *arXiv preprint arXiv:2204.00598* (2022).

ChatGPT3 on fractions :

Me: what is larger, two fourths or one half?

ChatGPT :

One half is larger than two fourths.

Two fourths can be written as 2/4, which is equal to 1/2. Therefore, one half is equal to two fourths.

Me: didn't you just contradict yourself?

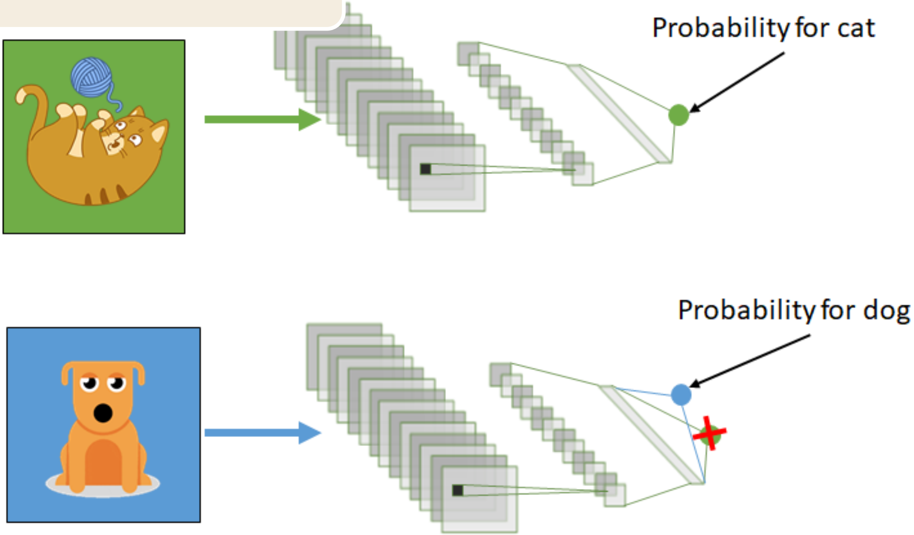
ChatGPT: I apologize, I made a mistake in my previous answer. Two fourths (2/4) is smaller than one half (1/2).

Me : Are you sure?

ChatGPT: Yes, I am sure. Two fourths (2/4) is equal to 1/2. Since 1/2 is larger than 2/4, it follows that one half is larger than two fourths.

Summary

Transfer learning



Backup slides

loss → Model

metrics → Model

hyperparameters → Model

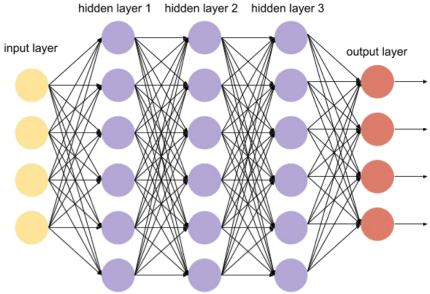
Goal ⟶ **Model** ⟶ **Training** ⟶ **Validation** ⟶ **Inference**

- **SL** prediction
- **UL** structure
- **RL** action

preprocessed data → Model

- ○ CV, train/test split
- ? under/overfitting

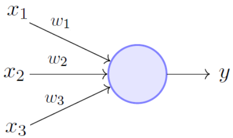
Neural network



$$h_t = f(W h_{t-1} + b)$$

$$a(x) = f_3(W_3 f_2(W_2 f_1(W_1 f_0(W_0 x + b_0) + b_1) + b_2) + b_3)$$

Calculating weights: slide 1



Perceptron Model (Minsky-Papert in 1969)

Source

$$Q(y_{model}, y_{target}) = Q(y(x, w, b), y_{target})$$

$$out = y = \sigma(sum) = \sigma\left(\sum_{i=1}^{3}(x_i \cdot w_i) + b\right)$$

$$\frac{\partial Q}{\partial w_i} = \frac{\partial Q}{\partial y} \cdot \frac{\partial y}{\partial \sigma} \cdot \frac{\partial \sigma}{\partial sum} \cdot \frac{\partial sum}{\partial w_i} = \frac{\partial Q}{\partial y} \cdot {\color{red}1} \cdot \frac{\partial \sigma}{\partial sum} \cdot \frac{\partial sum}{\partial w_i}$$

$$\frac{\partial \sigma}{\partial sum} = \sigma(sum) \cdot (1 - \sigma(sum))$$

Calculating weights: slide 2

$$Q\big(y(x,w,b),y_{target}\big)$$
$$Q=\big(y_{target}-y(x_1,w_1,b)\big)^2$$
$$out=y=(x\cdot w)+b$$
$$\frac{\partial Q}{\partial w_i}=\frac{\partial Q}{\partial y}\cdot\frac{\partial y}{\partial w_i}$$

Input: x = 1, w = 0.1 and b = 1

$y_{target}=2$ and learning rate: $\eta=0.1$

$$Q\big(y(x, w, b), y_{target}\big)$$
$$Q = \big(y_{target} - y(x_1, w_1, b)\big)^2$$
$$out = y = (x \cdot w) + b$$
$$\frac{\partial Q}{\partial w_i} = \frac{\partial Q}{\partial y} \cdot \frac{\partial y}{\partial w_i}$$

Input: x = 1, w = 0.1 and b = 1

$y_{target} = 2$ and learning rate: $\eta = 0.1$

How to find w_{new} and b_{new}?

$$\frac{\partial Q}{\partial y} = -2 \cdot (y_{target} - y)$$
$$\frac{\partial Q}{\partial w} = \frac{\partial Q}{\partial y} \cdot \frac{\partial y}{\partial w} = -2 \cdot x \cdot (y_{target} - y)$$
$$\frac{\partial Q}{\partial b} = \frac{\partial Q}{\partial y} \cdot \frac{\partial y}{\partial b} = -2 \cdot (y_{target} - y)$$
$$[w_{new}, b_{new}] \Rightarrow \theta_{i+1} = \theta_i - \eta \cdot \frac{\partial Q(\theta_i)}{\partial \theta_i}$$

$$Q\big(y(x, w, b), y_{target}\big)$$
$$Q = \big(y_{target} - y(x_1, w_1, b)\big)^2$$
$$out = y = (x \cdot w) + b$$
$$\frac{\partial Q}{\partial w_i} = \frac{\partial Q}{\partial y} \cdot \frac{\partial y}{\partial w_i}$$

Input: x = 1, w = 0.1 and b = 1

$y_{target} = 2$ and learning rate: $\eta = 0.1$

How to find w_{new} and b_{new}?

$$\frac{\partial Q}{\partial y} = -2 \cdot (y_{target} - y)$$
$$\frac{\partial Q}{\partial w} = \frac{\partial Q}{\partial y} \cdot \frac{\partial y}{\partial w} = -2 \cdot x \cdot (y_{target} - y)$$
$$\frac{\partial Q}{\partial b} = \frac{\partial Q}{\partial y} \cdot \frac{\partial y}{\partial b} = -2 \cdot (y_{target} - y)$$
$$[w_{new}, b_{new}] \Rightarrow \theta_{i+1} = \theta_i - \eta \cdot \frac{\partial Q(\theta_i)}{\partial \theta_i}$$

y	Q	$\frac{\partial Q}{\partial y}$	$\frac{\partial Q}{\partial w}$	$\frac{\partial Q}{\partial b}$	w_{new}	b_{new}
1.1	0.81	-1.8	-1.8	-1.8	0.28	1.18

Backup slides

Human brain